

# ARIA Claims Investigation Copilot

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## AI-Powered Fraud Detection & Investigation Platform

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Version 2.0

March 2026

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### Executive Summary

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ARIA (Advanced Risk Intelligence Assistant) Claims Investigation Copilot is an AI-powered platform that transforms how Special Investigation Units (SIU) detect and investigate insurance fraud. The system combines autonomous fraud detection with conversational AI, enabling investigators to uncover complex fraud schemes faster and with greater accuracy.

#### Key Business Outcomes:

- **85% faster investigation cycle time** — from days to hours
  - **2-3x increase in case throughput** per analyst
  - **40% improvement in fraud detection accuracy** through AI-powered pattern recognition
  - **Consistent, evidence-based reporting** for prosecution and recovery
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### Platform Overview

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#### What Is ARIA?

ARIA is an intelligent investigation assistant that works alongside your SIU analysts to:

1. **Monitor** incoming claims data for suspicious patterns
2. **Prioritize** high-risk cases requiring human review
3. **Investigate** using conversational AI that responds to analyst questions
4. **Discover** hidden fraud networks and relationships
5. **Document** findings in prosecution-ready dossiers

The platform supports two primary fraud investigation types:

- **Provider Fraud** — billing schemes, upcoding rings, phantom billing
  - **Disability Claims Fraud** — malingering, early filers, activity inconsistencies
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# Core Capabilities

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## 1. Case Queue Management

**What it does:** Automatically flags suspicious cases from overnight batch analysis and presents them in a prioritized investigation queue.

**Business Value:**

- Analysts start each day with clear priorities
- No manual case triaging required
- High-risk cases surface immediately

**Features:**

- Cases grouped by type (Provider Fraud, Disability Claims)
  - Priority scoring (HIGH, MEDIUM, LOW) based on risk indicators
  - Filter and sort by multiple criteria
  - Real-time case status tracking
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## 2. Conversational Investigation (Copilot Mode)

**What it does:** Analysts investigate cases by asking questions in natural language. The AI retrieves relevant data, performs analysis, and presents findings conversationally.

**Business Value:**

- Investigators work at the speed of thought
- No need to navigate multiple systems or run complex queries
- AI explains its reasoning, building investigator confidence
- Reduces training time for new analysts

**Example Conversation:**

Analyst: "Show me the timeline for this disability claim."

ARIA: "This claim was filed 4 months after policy purchase:

- Aug 15, 2025: Policy purchased
- Dec 1, 2025: Alleged disability onset
- Dec 22, 2025: Claim filed
- Jan 5, 2026: First medical visit

🚩 Red Flag: Policy age of 4 months is below the 6-month threshold commonly seen in suspicious early-filing patterns."

Analyst: "What am I missing? Could this be legitimate?"

ARIA: "Innocent explanations to consider:

1. Pre-existing condition that worsened unexpectedly
2. Occupational injury (check employer incident reports)
3. Legitimate short-tail disability (acute onset conditions)

The thin medical documentation (1 provider, 2 visits) supports concern, but request employment records before escalating."

## Investigation Tools by Case Type:

### Provider Fraud Tools:

- Entity profiling (billing patterns, peer comparisons)
- Claims analysis (CPT codes, frequency, cost outliers)
- Network detection (referral patterns, suspected rings)
- Referral history tracking
- Billing rule violation search

### Disability Claims Tools:

- Claim timeline reconstruction
- Medical records analysis
- Claimant history (prior claims, patterns)
- Policy detail review
- Inconsistency detection (activity vs. claimed limitations)
- Related claims search (provider patterns)

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## 3. Autonomous Investigation Mode (Auto Scan)

**What it does:** AI conducts a full investigation automatically, following best practices honed from thousands of prior cases. Runs in parallel while analysts work other cases.

### Business Value:

- Comprehensive analysis guaranteed on every case
- Catches connections human investigators might miss
- Scalable — run 10+ simultaneous investigations
- Perfect for initial case screening

#### Process:

1. **Scan** all claims data for anomalies
2. **Profile** high-risk entities
3. **Analyze** peer comparisons and statistical outliers
4. **Detect** fraud networks and rings
5. **Assess** evidence sufficiency
6. **Compile** dossier if evidence meets prosecution standards

**Typical Runtime:** 90-180 seconds per case

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## 4. Fraud Network Visualization

**What it does:** Interactive graph visualization of provider networks, referral relationships, and suspected fraud rings.

#### Business Value:

- Visual pattern recognition reveals schemes instantly
- Identify all co-conspirators in connected rings
- Evidence for conspiracy charges in prosecution
- Prioritize multi-entity takedowns

#### Features:

- Network density scoring
  - Anomaly highlighting (red nodes = high risk)
  - Interactive exploration (click to expand connections)
  - Export network diagrams for courtroom presentation
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## 5. Evidence-Based Dossier Generation

**What it does:** Automatically compiles investigation findings into structured, prosecution-ready reports.

#### Business Value:

- Consistent documentation across all cases
- Meets legal standards for evidence presentation
- Reduces manual report writing by 80%
- Includes precedent cases, billing rules, recovery estimates

#### Dossier Sections:

#### Provider Fraud Dossiers:

- Executive Summary with recommended actions
- Entities Involved (profiles and anomaly scores)
- Evidence Summary (statistical significance, peer comparisons)
- Statistical Analysis (z-scores, outlier metrics)
- Network Analysis (ring detection, connection density)
- Billing Rule Violations (CMS/OIG citations)
- Precedent Cases (similar schemes and outcomes)
- Recovery Estimate (collectability-adjusted projections)

#### Disability Claims Dossiers:

- Claim Timeline (chronological event sequence)
  - Medical Records Analysis (documentation quality, gaps, consistency)
  - Policy Analysis (age, benefit amounts, risk flags)
  - Inconsistencies & Red Flags (by severity)
  - Prior Claim History (patterns, benefit exhaustion)
  - Evidence Summary
  - Recommendations (escalation, IME, surveillance)
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## User Workflows

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### Workflow 1: Morning Case Review (Analyst Daily Routine)

1. **Open Case Queue** — 12 new cases flagged overnight
2. **Filter HIGH priority** — 5 cases require immediate attention
3. **Select Case DIS-2401** — Disability claim, early filer pattern
4. **Ask: "Timeline"** — AI shows 4-month policy age before claim
5. **Ask: "Medical records?"** — AI reveals thin documentation
6. **Ask: "What am I missing?"** — AI suggests innocent explanations
7. **Decision:** Request employer records before escalating
8. **Update Status:** "IN\_REVIEW" (case tracked in system)
9. **Move to next case**

**Time per case:** 3-5 minutes (vs. 30-45 minutes manual)

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## Workflow 2: Deep-Dive Provider Investigation

1. **Select Case NET-6610** — Provider fraud, 3.36σ billing anomaly
2. **Chat: "Profile this provider"** — AI shows specialty, volume, peer comparison
3. **Chat: "Network connections"** — AI reveals 4 connected entities
4. **Chat: "Ring detection"** — AI identifies suspected referral kickback scheme
5. **Click "Auto Scan"** — AI runs full autonomous investigation
6. **Review Live Activity** — Watch AI invoke tools, gather evidence
7. **Dossier Generated** — 12-page report with billing violations, precedent cases
8. **Decision:** Evidence sufficient — refer to OIG
9. **Export Dossier** — PDF for legal review

**Result:** \$3.4M estimated recovery, criminal referral package ready

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## Workflow 3: Weekly Fraud Pattern Analysis

1. **Network View** — Visualize all active fraud rings
2. **Filter: anomaly > 0.7, ring size ≥ 3** — 8 rings detected
3. **Select Ring #1** — 6 orthopedic surgeons, closed referral loop
4. **Investigate each provider** via Copilot chat
5. **Identify scheme:** Coordinated upcoding + referral kickbacks
6. **Generate dossiers** for all 6 entities
7. **Coordinate takedown** with legal team

**Result:** Multi-entity prosecution, 85% conviction rate on AI-flagged rings

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## Technical Requirements

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### Data Integration:

- Claims data (CPT codes, amounts, dates, providers, members)
- Provider master (NPIs, specialties, locations)
- Member demographics
- Policy information (for disability claims)
- Medical records (when available)

### System Requirements:

- Modern web browser (Chrome, Edge, Safari)
- Stable internet connection
- User authentication and role-based access control

## Security & Compliance:

- HIPAA-compliant data handling
  - Audit trails for all investigations
  - SOC 2 Type II certified infrastructure
  - End-to-end encryption for data in transit
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## Success Metrics

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### Operational Efficiency:

- 85% reduction in average case investigation time
- 3x increase in cases handled per analyst per week
- 60% reduction in backlog clearance time

### Detection Quality:

- 40% improvement in fraud detection accuracy
- 92% precision on HIGH-priority case flagging
- 15% increase in recovery amounts per case

### Analyst Satisfaction:

- 89% report feeling “more confident in findings”
  - 94% prefer AI-assisted over manual investigation
  - 78% reduction in repetitive research tasks
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## Implementation Approach

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### Phase 1: Pilot (4 weeks)

- Deploy to 3-5 senior analysts
- 50-100 case test dataset
- Weekly feedback sessions
- Measure baseline metrics

### Phase 2: Training & Rollout (6 weeks)

- Train all SIU analysts (2-hour sessions)
- Gradual case volume increase
- Side-by-side comparison with manual process
- Refine workflows based on feedback

### Phase 3: Full Production (Ongoing)

- All new cases routed through ARIA
- Monthly performance reviews
- Continuous model improvement
- Expand to additional fraud types

**Typical time to full adoption:** 10-12 weeks

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## Support & Maintenance

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### Included Services:

- 24/7 system uptime monitoring
- Monthly model retraining with latest fraud patterns
- Quarterly feature releases
- Dedicated support team (4-hour SLA)
- Training webinars and documentation updates
- Annual fraud trend reports

### Optional Add-Ons:

- Custom fraud pattern development
  - Integration with additional data sources
  - White-label branding
  - Advanced analytics dashboard
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## Pricing Model

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### SaaS Subscription:

- Per-analyst, per-month licensing
- Tiered pricing based on case volume
- Implementation and training included
- No hardware or infrastructure costs

### ROI Calculator:

- Average payback period: 3-6 months
  - Based on analyst productivity gains and increased recovery amounts
  - Contact sales for customized ROI analysis for your organization
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## Next Steps

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**Schedule a Demo:** See ARIA in action with your own claims data. We'll run a live investigation on a confirmed fraud case and show you the exact workflow your analysts will use.

**Start a Pilot:** Deploy ARIA to a small team for 30 days. Measure the impact on your key metrics. No long-term commitment required.

**Technical Integration:** Our solutions architects will work with your IT team to seamlessly integrate ARIA with your existing claims and case management systems.

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## Contact Information

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**Product Demo Requests:**

[demo.aria-fraud-intel.com](https://demo.aria-fraud-intel.com)

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*ARIA Claims Investigation Copilot — Empowering Human Expertise with AI Intelligence*

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